



Original research article

Risk in translation: Climate risk, social vulnerability, and clean energy support across the United States

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ABSTRACT

Public support for clean energy tends to follow perceived climate risk, but the relationship between objective climate hazards and that perception remains unexplored. Drawing on Yale Climate Opinion Maps, FEMA's National Risk Index, and 2024 election results across 3094 U.S. counties, we examine geographic co-variation among objective climate risk, perceived risk, social vulnerability, political context, and support for a clean energy transition. We found that support is strongly spatially clustered, closely tracking regional patterns of risk perception and political identity. The relationship between objective hazards and perceived risk reverses across the political spectrum: in liberal-leaning counties, greater hazard exposure coincides with higher perceived risk, whereas in conservative-leaning counties, it coincides with lower perceived risk. Perceived climate risk is the strongest county-level correlate of clean energy support, but this association is weaker in conservative counties than in liberal ones. Social vulnerability is positively associated with support in conservative counties, and negatively in moderate and liberal ones. The same objective risk and vulnerability thus translate into markedly different risk perceptions and energy preferences across political contexts, and strategies that ignore this asymmetry are unlikely to build durable support.

1. Introduction

From wildfires in California to hurricanes in the Gulf, the United States faces escalating climate risks. Yet turning these tangible threats into public support for clean energy involves a complex interplay of perception, politics, and local vulnerability. Urgent action is needed to reduce greenhouse gas emissions and accelerate renewable energy adoption [1–3]. A large-scale transition to renewable energy is widely recognized as a central strategy for mitigating global warming, reducing reliance on fossil fuels, and fostering sustainable economic growth [4]. As a major actor in global climate dynamics, the United States (U.S.) plays a pivotal role in advancing this transition through domestic policies and by setting international benchmarks. The United States, as a significant player in global climate dynamics, occupies a position where domestic choices may carry international weight, with domestic policy trends often reflected in broader global trajectories. Understanding the factors associated with American support for clean energy is therefore critical, not only for national progress but for the broader ambition of coordinated global climate action [5–7].

Recent extreme weather highlights the urgency of action. In 2024, the United States recorded its warmest year in 130 years [8] and the second-highest number of billion-dollar disasters, with damages exceeding \$182.7 billion and expected to rise. Hurricanes Helene and Milton alone caused over \$100 billion in losses, along with widespread fatalities and displacement. These escalating human and economic impacts highlight the need for accelerated emissions reductions to reverse current trends and strengthen long-term resilience.

Despite broad scientific consensus on the urgency of decarbonization [3,9], public support for renewable energy policies remains inconsistent across the U.S. [7,10–12]. Variations in support often reflect political ideology, perceptions of environmental risk, or differential exposure to climate-related impacts [1,13]. Climate change risks are typically conceptualized as three dimensions: objective measures of community-level expected annual loss from severe weather events, indicators of social vulnerability and subjective risk perceptions [14,15]. Yet, research shows that the reality of physical climate risks and perceived risks frequently diverge [16]. Importantly, the intersections among objective climate change risks, subjective perceptions and public

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preferences for clean energy transitions and solutions remain understudied [15].

To address these gaps, this study investigates how climate risks, perceived climate risks, social vulnerability, and political orientation are associated with public preferences on energy transition in the U.S. Specifically, we ask (1) how is objective climate risk associated with perceived climate risk across U.S. counties, and how is this relationship conditioned by political orientation? and (2) how are objective climate risk, social vulnerability, and estimated perceived climate risk associated with county-level support for the clean energy transition? By integrating environmental, psychological, and political predictors, within a spatially explicit, multilevel framework, this study contributes evidence on how these factors co-vary across U.S. counties and offers a more contextual basis for designing communication and policy strategies that account for the political and social conditions under which support for sustainable energy transitions takes shape.

2. Literature review

2.1. Role of clean energy transition

Greenhouse gas emissions (GHG), particularly carbon dioxide and methane released from the combustion of coal, oil, and natural gas, are the primary drivers of global warming, which in turn intensifies extreme weather events, accelerates sea level rise, and disrupts ecosystem [17,18]. Transitioning to clean energy has thus emerged as a critical pathway for mitigating climate change, improving public health, strengthening energy security, and fostering long-term economic and environmental resilience in the United States [4,19].

The changes in energy production and consumption of the last several decades led to a slight decrease in the U.S. emissions of CO₂, but total global greenhouse gas emissions continue to increase [3]. The U.S. National Climate Assessment underscores that climate-related disasters have intensified in recent decades, producing billions of dollars in damages annually and overwhelming emergency response systems [18]. Physical climate risks, including wildfires, hurricanes, droughts, and floods, are escalating across regions, with cascading effects on public health, food and water security, infrastructure, and economic stability [20]. The energy sector is also a leading source of harmful co-pollutants, driving smog formation, acid rain, and adverse health outcomes [21]. These environmental and health burdens fall disproportionately on low-income communities and communities of color situated near power plants, refineries, and industrial corridors [22]. Advocates, therefore, stress the need for equitable access to the benefits of clean energy, including affordable electricity, cleaner air, and meaningful participation in energy decision-making, particularly for historically marginalized populations [23].

Economically, the clean energy transition presents significant opportunities for job creation and innovation. According to the U.S. Department of Energy [24], employment in renewable energy sectors, especially solar and wind, has grown faster than in fossil fuel industries. Investments in clean energy infrastructure also stimulate local economic development, particularly in rural and economically distressed regions [25]. Moreover, by diversifying the energy supply and reducing dependence on foreign oil, clean energy enhances national security [26].

Despite broad recognition of the need to mitigate climate change, climate policy implementation and compliance are strongly shaped by environmental awareness [26], trust in key stakeholders [27,28], and the level of public support [10–12]. A 2023 Pew Research Center survey found that 67% of Americans prioritize alternative energy sources over fossil fuels, and large majorities support measures such as carbon taxes and clean energy tax credits [23]. However, support remains uneven, shaped by the interplay of political ideology, regional economic dependence on fossil fuels, and climate-related risks [7,29].

Public support for renewable energy is closely linked to climate risk perceptions: individuals who view climate change as a serious and

personal threat are significantly more likely to endorse clean energy policies [13,30]. Yet, evidence suggests that even in the aftermath of extreme weather events, public attitudes shift only gradually [31]. Although substantial scholarship has examined the influence of objective and perceived risks on policy preferences, how these risks jointly co-vary with political context and social vulnerability across the U.S. landscape remains poorly understood [16].

2.2. Reality vs. perceptions: objective and perceived climate risks

Risks associated with global warming and environmental degradation are widely viewed as a central driver of recognition and response to climate change. Yet, risks are not simply objective facts; they can be amplified, minimized, dramatized, or redefined through social processes, rendering them open to interpretation and construction [32,33]. Risks are typically conceptualized along two dimensions: objective measures, grounded in community-level vulnerability and exposure, and subjective perceptions, shaped by personal experience, ideology, and contextual influences [14,15,34].

Objective climate risks capture the short- and long-term impacts of extreme weather events on people, infrastructure, and ecosystems [5]. Climate risk assessments quantify exposure to hazards such as droughts, flooding, and extreme heat, often using expected annual loss estimates from federal agencies such as the Federal Emergency Management Agency (FEMA), the U.S. Army Corps of Engineers, and the U.S. Department of Agriculture [35].

While frequency and intensity of such events can shape public attitudes toward climate change, findings remain mixed. Some studies demonstrate that local weather conditions are internalized by individuals and significantly influence climate attitudes [13,34,36]. Severe local events have been linked to greater concern about climate change and increased support for mitigation policies [37–39].

Subjective risk perceptions, however, do not always align with objective measures, such as recorded weather patterns, the frequency of disasters, or estimated annual losses [16]. Evidence reveals weaker or inconsistent relationships [40,41]. Studies show that the perceived likelihood or severity of a disaster often bears little relation to empirical climate risk levels [41,42]. Reviews by Howe [43] and Sambrook et al. [44] underscore these inconsistencies, noting that the mechanisms linking weather experiences to climate perceptions likely involve multiple mediating factors. Risk perceptions are shaped by broader social and institutional contexts, including political ideology and lived experiences [15]. For instance, Hitchner et al. [45] found that rural forest landowners' perceptions of climate risk, and their support for renewable energy, were influenced by a blend of ideology, occupational context, and personal observations. Moreover, risk perceptions are dynamic rather than fixed, shifting in response to new experiences, evolving information, or targeted communication strategies [40].

Risk perception also extends beyond individual cognition to encompass collective processes, shaped by media framing, narratives, and social networks. Group dynamics and interpretive frames can amplify or attenuate perceived risks [46]. These dynamics influence how individuals process information, update their understanding of risks, and adjust their behaviors, including their support for clean energy transitions [47].

2.3. Social vulnerability and climate risks beliefs

Socioeconomic disparities further complicate public support for mitigation. Contrary to assumptions that climate risks affect all populations equally, evidence shows that vulnerable groups face greater health and livelihood impacts [46]. Contributing factors include access to social networks and interplay of socioeconomic and political contexts shaping vulnerability. Support for climate policies also varies by vulnerability (e.g., single mothers, older adults, migrants), depending on whether people prioritize immediate needs, impact severity, and

perceived policy benefits [48].

Vulnerable communities may overestimate the negative consequences of clean energy infrastructure, dampening support [10] and often resist policies like carbon taxes due to financial concerns [49]. In some high-risk regions, climate skepticism has reduced backing for renewables energy [50]. Conversely, studies also find that vulnerable communities often express greater concern about climate change and stronger support for protective policies [51], reflecting growing recognition of energy justice and unequal environmental harms [52].

2.4. Political orientation and climate policy preferences

Political competition and partisanship are central determinants of public attitudes toward climate action and clean energy policy [53–56]. A review by Struthers et al. [29] highlights that although support for clean energy has been increasing among conservatives, Democrats consistently express stronger backing for renewable energy than Republicans. Conservatives remain less likely to accept the scientific consensus on climate risks, and their ideological orientations are often associated with lower risk perceptions and reduced support for mitigation [40]. This contributes to a persistent misalignment between objective and perceived risks, undermining adaptation planning across regions [16].

Political divides remain sharp: democrats and liberals generally support climate mitigation and clean energy policies, while conservative Republicans are more skeptical, particularly toward government-led efforts [57–59]. For instance, 87% of Democrats favor expanding wind, solar, and hydrogen energy, compared to 60% of Republicans who prioritize fossil fuel development [59]. Conservative skepticism is often rooted in concerns about excessive government regulation and its potential to disrupt economic interests and lifestyles [1]. Furthermore, in regions where local economies and public services are heavily reliant on fossil fuel industries, changes in energy infrastructure are often perceived not just as policy shifts but as direct threats to community livelihoods [60,61]. Such dynamics amplify resistance and contribute to political reluctance in advancing clean energy initiatives [10].

Beyond these direct effects, political orientation also shapes climate attitudes indirectly. Studies find that political ideology moderates the relationship between climate risk perception and support for renewable energy. For example, Hwang et al. [62] show that individuals' acceptance of renewable energy is conditioned by their political ideology, even when risk perceptions are similar. These findings illustrate how political ideology influences both the interpretation of climate risks and the perceived legitimacy of proposed solutions [63]. Overall, partisan polarization remains a major obstacle to the renewable energy transition. Deep ideological divides continue to constrain policy adoption, weaken public consensus, and threaten progress toward national climate goals [54].

In sum, while national surveys suggest growing support for renewable energy, policy implementation remains constrained by political polarization, divergent risk perceptions, economic self-interest, and embedded social vulnerabilities. Understanding how these factors interact is essential for designing climate and clean energy policies that are both effective and equitable.

2.5. Integrated theoretical framework

Our analysis is informed by three lines of scholarship that emphasize how social and political context shape responses to environmental risks. Research on risk governance [42,64,65] treats environmental risks as socially embedded phenomena whose recognition depends on institutions, trust, and the credibility of those communicating risk information, suggesting that objective hazards do not translate automatically into public concern or policy support. Environmental and energy justice scholarship [22,52,66] highlights that exposure and the capacity to respond to environmental risks are unequally distributed, and that social

vulnerability conditions how communities relate to climate and energy policy. Inglehart's Objective Problem–Subjective Value hypothesis [67] further links objective conditions to public attitudes through the material and political contexts in which exposure is interpreted, suggesting that the association between objective risk and policy support varies across communities rather than holding uniformly. In the U.S., political ideology is closely associated with how climate information is received and how risks are evaluated [68,69]. While Beck's [33] earlier work establish risk as a defining feature of late modernity, the empirically grounded risk governance tradition developed by Rosa, Renn, McCright, and colleagues [64] more directly informs the present analysis.

These perspectives motivate a two-stage descriptive framework. We expect objective climate risk to be associated with perceived risk, but with this association conditioned by political context and social vulnerability. We further expect perceived risk to align more closely with support for the clean energy transition than objective risk does. Because the variables available at the county level are constructed indicators rather than directly measured constructs, we treat this framework as describing geographic co-variation patterns rather than individual-level causal mechanisms.

2.6. Research questions and hypotheses

This study examines how objective local climate exposure is associated with county-level climate risk perceptions and, in turn, support for clean energy policies. We expect that higher levels of climate risk are associated with higher risk perceptions and stronger support for clean energy transitions. However, perceptions of climate risk are shaped by social and political context and may not reflect objective exposure uniformly across counties.

Political orientation is expected to condition how objective risks are translated into perceived risks. In more conservative counties, the relationship between exposure and perception may be weaker; in more liberal contexts, this relationship is expected to be stronger. Building on prior research, we examine this moderating role within a two-stage framework linking objective risk, perceived risk, and support for clean energy policies.

Because perceived risk and policy support are derived from modeled county-level estimates, the analysis focuses on associations between estimated quantities rather than individual-level causal effects. Accordingly, we examine patterns of geographic alignment among objective climate exposure, estimated perceptions, political context, and policy support. We specifically test the following research questions (RQs):

RQ1: How is objective climate risk associated with perceived climate risk across U.S. counties, and how is this relationship conditioned by political orientation?

RQ2: How are objective climate risk, social vulnerability, and estimated perceived climate risk associated with county-level support for the clean energy transition? The list of hypotheses (Hs) is presented in the appendices Table A.

3. Method

3.1. Data sources

This study uses two major publicly available data sets describing U.S. public perceptions of climate risk from the 2024 wave of Yale Climate Opinion Maps (YCOM) developed by the Yale Program on Climate Change Communication and data from the U.S. Federal Emergency Management Agency (FEMA) National Risk Index (NRI), along with county-level political voting patterns from the 2024 presidential election.

3.1.1. Yale Climate Opinion Maps

The YCOM 2024 wave by the Yale Program on Climate Change

Communication and the George Mason University Center for Climate Change Communication offers insights into public perceptions of climate change across various demographics and regions. The dataset provides estimates generated through a statistical approach that integrates multilevel regression with post-stratification, leveraging demographic and geographic predictors from a series of nationally representative surveys conducted annually since 2014. For the estimation, the model includes four geography-level covariates: percent of people who drive alone to work, CO2 emissions per capita, percent of same-sex households, and most recent elections.

For the 2024 dataset, estimates include 95% confidence intervals derived from 99 bootstrap simulations. These intervals suggest an expected margin of error of approximately ± 8 percentage points for county-level data. These ranges also encompass the baseline sampling error from the original surveys, typically around ± 3 percentage points, reflecting a high degree of model reliability for localized public opinion estimates [70].

3.1.2. National Risk Index

The U.S. Federal Emergency Management Agency (FEMA) National Risk Index (NRI) database provides comprehensive data regarding community-level climate risk vulnerabilities and resilience, which helps to identify regions most at risk from climate hazards [71]. In the Risk Index, risk is defined as the potential for negative impacts because of a natural hazard. The NRI incorporates three main components to assess the risk of natural hazards for communities across the United States: Expected Annual Loss (EAL), Social Vulnerability and Community Resilience [71]. The NRI calculates risk using the following equation:

$$\text{Risk} = \text{Expected Annual Loss} \times (\text{Social Vulnerability} / \text{Community Resilience})$$

The NRI data set was initially released in October 2020, with subsequent updates in August and November 2021 (v.1.18) and the current release in March 2023 (v.1.19). The March 2023 version updated the dataset to 2021 Census boundaries and population for counties [71].

3.2. Measurements

Table B lists all the measurements of independent and dependent variables.

3.2.1. Dependent variable: support for clean energy transition

Support for the clean energy transition was measured using seven items from the YCOM survey that capture agreement with or support for a range of energy-related policies. Each item originally offered four response categories. For analytic purposes, responses were collapsed to reflect the percentage of county residents who indicated that they agree or somewhat agree (or support or somewhat support) with the following statements:

1. Requiring fossil fuel companies to pay a carbon tax, with the revenue used to reduce other taxes (e.g., income tax) by an equal amount.
2. Making the development of clean energy sources a high or very high priority for the president and Congress.
3. Regulating carbon dioxide as a pollutant.
4. Expanding offshore drilling for oil and natural gas off the U.S. coast.
5. Drilling for oil in the Arctic National Wildlife Refuge.
6. Funding research into renewable energy sources.
7. Providing tax rebates for individuals who purchase energy-efficient vehicles or solar panels.

For items 4 and 5, which measure support for fossil fuel expansion, responses were reverse-coded so that higher values consistently indicate stronger support for the clean energy transition. Descriptive statistics (means and standard deviations) for all items are presented in appendices Fig. A.1.

To estimate county-level support for clean energy transition, we conducted an exploratory factor analysis and examined the latent structure of the Support for Clean Energy Transition Scale. The two items measuring support for oil drilling were reverse-coded to reflect the percentage of respondents who opposed such activities, ensuring consistency in the direction of all items. Preliminary tests confirmed the data's suitability for factor analysis: the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy was 0.898, and Bartlett's Test of Sphericity was significant ($p < .001$). Factors were extracted using Principal Axis Factoring, with retention determined by the eigenvalue > 1 criterion. Results indicated a single dominant factor that explained 89.36% of the total variance, interpreted as Support for Clean Energy Transition. All items loaded strongly on this factor (range = 0.898–0.963), and no items were removed. The scale demonstrated excellent internal consistency, with a Cronbach's alpha of 0.976.

3.2.2. Perceived risks

Subjective perceptions of climate risks were measured using six items from the YCOM Risk Perceptions module. For each county, we calculated the percentage of residents who reported: (i) personally, experiencing the effects of global warming, (ii) believing that global warming will cause a moderate or great amount of harm to plants and animal species, (iii) believing that it will harm future generations, (iv) believing that it will harm people in developing countries, (v) believing that it will harm people in the US, and (vi) believing that it will harm them personally.

Across U.S. counties, approximately 40% of residents reported having personally experienced the effects of global warming and expected personal harm from climate change. Over 55% believed that climate change would cause harm to the U.S., and about 65% expressed concern for future generations. Among these dimensions, concern for future generations had the highest average level, followed by concern for plants and animals, while perceptions of personal risk were consistently the lowest across all political groups. Political orientation was strongly associated with these views: average levels of perceived climate risks were lower in Republican-leaning counties and higher in Democratic-leaning counties (Fig. A.2). To investigate the latent structure of perceived climate risks, we conducted an exploratory factor analysis of the six items. Results supported a unidimensional construct of perceived risk, with excellent internal consistency (Cronbach's $\alpha = 0.978$).

3.2.3. Measure of objective climate risk

Objective climate risk was measured using the Expected Annual Loss (EAL) indicator from FEMA's NRI. The EAL estimates the anticipated annual economic damage from natural hazards, integrating three components: exposure, annualized frequency, and historic loss ratio. Exposure refers to the assets potentially impacted by hazards (e.g., buildings, population, agriculture). Annualized frequency represents the probability of a hazard occurring each year. Historic loss ratio reflects the expected proportion of damage relative to exposed assets. EAL is expressed in U.S. dollars and is calculated for 18 natural hazards [71].

3.2.4. Social vulnerability

Social vulnerability was measured using the Social Vulnerability Index (SVI) from the U.S. CDC's Agency for Toxic Substances and Disease Registry (ATSDR), as integrated into the FEMA NRI. The SVI provides a location-specific assessment of communities' susceptibility to disproportionate adverse outcomes from natural hazards, including death, injury, property loss, and livelihood disruption. The index is based on 16 socioeconomic variables that capture reduced capacity to prepare for, respond to, and recover from hazards. Higher scores indicate greater social vulnerability and, consequently, higher risk [71].

3.2.5. Political orientation

County-level political orientation was operationalized using the percentage of votes cast for Democratic and Republican candidates in

the 2024 U.S. presidential election [72]. This variable was scaled from 0 to 100, with 0 representing counties that voted entirely Republican and 100 representing counties that voted entirely Democratic. The 0–100 scale was chosen for comparability with other study variables, which are reported on similar percentage-based scales. To capture categorical variation, the variable was mean-centered, and counties were grouped relative to one standard deviation ($SD = 15.62$) from the mean: Conservative-leaning counties: $>1 SD$ below the mean ($n = 428$; 13.9%), Moderate counties: within $\pm 1 SD$ of the mean ($n = 2148$; 69.3%), Liberal-leaning counties: $>1 SD$ above the mean ($n = 518$; 16.7%).

Because the 2024 election results were highly skewed toward the Republican candidate, this grouping approach (mean $\pm SD$) was used instead of tertiles to better capture distributional differences across political contexts.

3.3. Test variables and implications of YCOM estimation

YCOM variables (perceived risk and policy support) are county-level estimates based on survey data combined with demographic and political predictors and are partly functions of overlapping county-level characteristics. While the YCOM estimates have been externally validated against independent survey data [73], providing a basis for treating them as meaningful measures of county-level opinion, there remains a possibility of mechanical correlation: with observed relationships reflecting shared estimation rather than independent empirical associations. To mitigate this concern, our approach follows established practice for addressing potential confounding through covariate inclusion [74]. We incorporate independently sourced measures of the key geographic covariates used in the YCOM model: percentage of commuters who drive alone to work and the percentage of same-sex households from the U.S. Census, CO₂ emissions per capita from the U.S. Energy Information Administration (EIA), and political context based on the 2024 presidential election. These variables approximate the constructs used in the YCOM estimation while relying on independent data sources and updated information. The inclusion of all four YCOM estimation covariates in Model 5 represents the most direct test available within the current data structure, and the stability of core coefficients across this specification provides partial reassurance that results are not driven entirely by shared inputs. We acknowledge, however, that this cannot fully resolve the reviewer's concern, and we have been explicit throughout the revised manuscript that findings should be interpreted as descriptive of geographic co-variation patterns rather than independent causal effects.

In addition, we estimate alternative model specifications excluding perceived climate risk to assess whether relationships between objective climate risk and policy support persist independently of the modeled perception variable. These steps allow us to evaluate the robustness of our findings and reduce concerns that results are driven solely by the internal structure of the YCOM estimates.

3.4. Data preparation and analytical approach

We construct a county-level dataset by merging five primary sources: (1) Yale Climate Opinion Maps (YCOM), which provide modeled estimates of perceived climate risk and support for clean energy policies; (2) the National Risk Index (NRI), which includes Expected Annual Loss (EAL) and the Social Vulnerability Index (SVI); (3) county-level political voting patterns from the 2024 U.S. presidential election, (4) the percentages of people drive alone and same sex households (US Census) and the CO₂ per capita (EIA). The data were analyzed using IBM SPSS Statistics (Version 29) [75]. The analytical sample excludes Alaska and Connecticut and several counties from other states due to missing data or boundary inconsistencies across the datasets integrated in this study. We excluded Kalawao County in Hawaii, Blaine and McPherson Counties in Nebraska, Concho, Kenedy, King, and Loving Counties in Texas, and Catron County in New Mexico due to missing entries. Because

the analysis relies on county-level data from multiple independent sources, reconciling divergent geographic boundary definitions across datasets was not feasible. Alaska presents well-documented challenges for county-level analysis due to its borough and census area structure, which does not align consistently with county-level estimates in other datasets. Connecticut recently transitioned from a county-based administrative structure to planning regions, creating boundary mismatches. All other U.S. states and the District of Columbia are included, yielding a final analytical sample of 3094 counties.

Scales for perceived risk and policy support were constructed. Descriptive analysis and correlations are presented in Table C. Predictor variables were mean-centered and interaction terms computed.

3.4.1. Approach to analysis

The analysis proceeds in two main stages corresponding to the research questions. For RQ1, we examine how objective climate risk is associated with estimated perceived risk using sequential block-entry regression models that include political orientation and an interaction term to assess moderation. Following Brambor, Clark, and Golder [76], the interaction was specified as the product of mean-centered objective climate risk and mean-centered political orientation, with both constitutive terms retained. Because the substantively meaningful quantity in an interaction model is the marginal effect of the focal predictor across the range of the moderator rather than the interaction coefficient alone [76,77], we plot marginal effects of objective climate risk on perceived risk across the observed range of political orientation, with 95% confidence intervals, and identify the Johnson-Neyman threshold. Stratified models by political context (conservative, moderate, liberal) are reported as complementary descriptive results.

For RQ2, we examine how estimated perceived risk, objective climate risk, and social vulnerability are associated with support for the clean energy transition. We estimate sequential block-entry regression models to assess baseline relationships and compare specifications with and without perceived risk, followed by linear mixed-effects models with counties nested within states to account for geographic clustering. The interaction between objective climate risk and political orientation is specified following the same approach as in RQ1.

Next, we assess geographic clustering in support using global Moran's I and Local Indicators of Spatial Association (LISA). Data preparation and geographic visualization were conducted in QGIS [78] while spatial autocorrelation analyses, including Moran's I and Local Indicators of Spatial Association (LISA), were performed using GeoDa [79]. To evaluate model performance, we examine spatial autocorrelation in residuals. Finally, standard diagnostic tests (residual plots, Q-Q plots, and leverage statistics) are used to assess model assumptions.

4. Results

4.1. Descriptive statistics

Nationally, most counties expressed support for clean energy transition ($M = 61.78\%$) and even in the counties with the lowest support, nearly half of residents (48.98%) endorsed the transition. Figure B presents county-level support for the clean energy transition. The results of Pearson correlations (Table C) showed that all variables were significantly positively correlated. Counties with higher climate risk had a larger portion of the population who recognized risks from global warming ($r = 0.42$, $p > .001$), and a higher prevalence of Democratic voters ($r = 0.41$, $p > .001$). Counties with higher social vulnerability levels had higher objective climate risks ($r = 0.195$, $p > .001$), were more likely to recognize these risks ($r = 0.107$, $p > .001$) and to support clean energy transition policies ($r = 0.103$, $p > .001$).

Political orientation was highly associated with both estimated perceived climate risk ($r = 0.840$, $p < .001$) and support for the clean energy transition ($r = 0.899$, $p < .001$), and perceived risk and support were themselves strongly correlated ($r = 0.916$, $p < .001$). These strong

relationships reflect both the central role of political context in shaping climate attitudes in the U.S. and the fact that these variables are derived from YCOM estimates using overlapping inputs. Accordingly, the findings should be interpreted as patterns of alignment among closely related constructs rather than fully independent effects.

4.2. Climate risk perception (RQ1)

Next, we analyzed how objective climate risk is associated with perceived climate risk across U.S. counties (RQ1). Table 1 reports the regression results examining the association between objective climate risk, social vulnerability, political orientation, and estimated perceived climate risk. Social vulnerability showed a weak and inconsistent association with perceived climate risk. Objective climate risk was positively associated with perceived risk ($\beta = 0.417, p < .001$) in the baseline model ($R^2 = 0.179, F(2, 3092) = 338.823, p < .001$). The inclusion of political orientation, used in the YCOM estimation, substantially improved model fit $R^2 = 0.714, F(3, 3091) = 2589.369, p < .001$ and attenuated the contribution of climate risk ($\beta = 0.101, p < .001$). The interaction between objective climate risk and political orientation was positive and significant ($\beta = 0.142, p < .001$), indicating that the relationship between objective risk and perceived risk varied across political contexts.

Figure C presents the marginal effect of objective climate risk on perceived risk across the observed range of political orientation, with 95% confidence intervals. The effect rises monotonically from about -0.04 at the Republican end of the distribution to about 0.13 at the Democratic end, crossing zero at approximately 23% Democratic vote share (the Johnson–Neyman threshold). Below this threshold, higher objective climate risk is significantly associated with lower perceived risk; above it, the association is significantly positive. This continuous pattern is consistent with the stratified subgroup results in Table 2 and shows that the moderation operates across the political distribution, not only at the conservative–moderate–liberal cutpoints used in the categorical analyses.

To further interpret interaction effect, we estimated stratified models by political orientation. The regression models were significant across all three groups (Models 2.1–2.3, Table 2). The explanatory power of the models was substantially higher in liberal counties ($R^2 = 0.181, p < .001$) compared to conservative ($R^2 = 0.016, p = .033$) and moderate counties ($R^2 = 0.095, p < .001$) suggesting that here objective conditions shape perceptions more consistently. The association between climate risk and perceived risk was strongest in liberal counties ($\beta = 0.359, p < .001$), followed by moderate counties ($\beta = 0.299, p < .001$). Notably, in Conservative counties, climate risk showed a negative coefficient ($\beta = -0.129, p = .01$), indicating that higher objective climate risk was associated with lower perceived risk. Social vulnerability was significant predictor only in liberal counties ($\beta = -0.200, p < .001$). However, the model explains only a small fraction of variance in this subgroup.

Table 1
Models 1–3 regression coefficients on perceived climate risks.

Predictors	Model 1	Model 2	Model 3
	β	β	β
Social vulnerability	$-0.028, p = .095$	$-0.030, p = .396$	$-0.013, p = .17$
Climate risk	0.417^{***}	0.101^{***}	0.079^{***}
Political orientation		0.804^{***}	0.786^{***}
Interaction ^a			0.142^{***}
R^2	0.179^{***}	0.714^{***}	0.733^{***}
F	$(2, 3092) = 338.823$	$(3, 3091) = 2589.369$	$(4, 3090) = 2135.817$

β standardized coefficient.

*** $p < .001$.

^a Interaction effect of political orientation and climate risk.

Table 2

Regression coefficients of Models 2.1–2.3 on perceived climate risks for political groups.^a

Coefficients	Conservative counties	Moderate counties	Liberal counties
Climate risk, β	-0.129^*	0.299^{***}	0.359^{***}
Social vulnerability, β	$0.01, p = .846^{**}$	$0.032, p = .128$	-0.200^{***}
R^2	$0.016^*, p = .033$	0.095^{***}	0.181^{***}
F	$(2, 426) = 3.449$	$(2, 2146) = 112.122$	$(2, 516) = 57.041$

β standardized coefficient.

*** $< .001$.

** $< .01$.

* $< .05$.

^a Conservative-leaning counties - 1 SD below the mean, moderate - within ± 1 SD of the mean, and liberal-leaning 1 SD above the mean.

Figure D shows the relationship between objective and perceived risk across the groups.

4.3. Support for energy transition policies (RQ2)

Having examined the relationship between objective climate risk and perceived risk, we next analyzed how these factors relate to support for the clean energy transition. Table 3 presents sequential block-entry regression models examining the association between objective climate risk, social vulnerability, political orientation, and perceived climate risk with support for the clean energy transition.

Model 1 included only objective climate risk and social vulnerability. It was statistically significant ($F(2, 3092) = 377.778, p < .001$) and explained a modest proportion of the variance ($R^2 = 0.196$). Objective climate risk was positively associated with support for clean energy transition ($\beta = 0.439, p < .001$), while social vulnerability was not statistically significant ($\beta = 0.017, p = .295$).

Model 2 added political orientation and showed a substantial increase in explanatory power ($R^2 = 0.819, F(3, 3091) = 4679.294, p < .001$). Political orientation emerged as the dominant predictor ($\beta = 0.866, p < .001$), partly expected given construct overlap. The effect of objective climate risk was reduced but remained significant ($\beta = 0.100, p < .001$), and social vulnerability became negatively associated with support ($\beta = -0.044, p < .001$).

Model 3 introduced estimated perceived climate risk and further improved model fit ($R^2 = 0.898, F(4, 3090) = 6805.235, p < .001$). The inclusion of perceived risk substantially attenuated the association between objective climate risk and support, indicating that perceived risk accounts for a significant share of this relationship ($\beta = 0.529, p < .001$). Political orientation remained significant ($\beta = 0.440, p < .001$), while the effect of objective climate risk was further attenuated ($\beta = 0.045, p < .001$). Social vulnerability retained a small negative effect ($\beta = -0.028, p < .001$). Model 4 added the interaction between objective climate risk and political orientation. The model remained significant ($F(5, 3089) = 5442.472, p < .001$), but the interaction term was not statistically significant.

In Model 5, to assess the sensitivity of the main results to shared model inputs, we included three geographic covariates used in the YCOM estimation process (same-sex households, drive-alone commuting, and CO₂ emissions per capita) alongside political orientation. The main coefficients were modestly attenuated but substantively stable: perceived risk ($\beta = 0.511, p < .001$) and political orientation ($\beta = 0.422, p < .001$) remained the strongest predictors, while objective climate risk retained a small positive association ($\beta = 0.018, p < .001$). This stability suggested the core relationships were not driven entirely by shared inputs, though the degree to which they reflect independent substantive associations or partially correlated estimation procedures cannot be fully determined from these data. The findings should

Table 3

Models 1–5 regression coefficients on support for the clean energy transition.

Predictors	Model 1	Model 2	Model 3	Model 4	Model 5
	β	β	β	β	β
Climate risk	0.439***	0.100***	0.045***	0.045***	0.018***
Social vulnerability	0.017, $p = .295$	−0.044***	−0.028***	−0.028***	−0.035***
Political orientation		0.866***	0.440***	0.441***	0.422***
Perceived risk			0.529***	0.528***	0.511***
Interaction ^a				0.001, $p = .881$	0.011, $p = .053$
Same-sex households					0.028***
Drive-alone					−0.044***
CO ₂ per capita					−0.103***
R ²	0.196***	0.819***	0.898***	0.898***	0.910***
F	(2, 3092) = 377.778	(3, 3091) = 4679.294	(4, 3090) = 6805.235	(5, 3089) = 5442.472	(8, 3086) = 3923.693

 β standardized coefficient.*** $p < .001$.^a Interaction effect of political orientation and climate risk.

therefore be interpreted as descriptive of co-variation patterns among county-level estimated attitudes, objective risk, and political context, rather than as evidence of independent causal effects. The interaction between climate risk and political orientation became marginally significant ($\beta = 0.011$, $p = .053$) in this model. Variance inflation factors (VIFs) across all the models indicated no evidence of problematic multicollinearity. Spatial dependence was partially accounted for through multilevel modeling, with residual analyses reported in Section 4.4.

To account for the hierarchical structure of the data, we estimated linear mixed-effects models with counties nested within states and random intercepts for states, presented in Table D. The intraclass correlation coefficient ($ICC = 0.416$) indicated that approximately 41.6% of the variance in support for the clean energy transition was attributable to differences between states, confirming substantial geographic structure in the data. In the multilevel specification (Table D), the interaction between objective climate risk and political orientation showed a small negative coefficient ($B = -0.000096$, $p = .082$). The magnitude is very small in absolute terms, and statistical detectability is borderline by conventional thresholds. We interpret this as suggestive of weak moderation that becomes detectable only after accounting for state-level clustering, consistent with the pattern observed in RQ1, where the relationship between objective and perceived risk varied substantively by political context. Given the small effect size, we do not draw strong substantive conclusions from this interaction in the support for clean energy transition model.

Variance component estimates further confirmed geographic heterogeneity, with state-level variance estimated at 1.137 ($SE = 0.237$) and residual variance at 1.597 ($SE = 0.041$). Model fit statistics showed strong explanatory power (marginal $R^2 = 0.907$, conditional $R^2 = 0.945$), indicating that the included variables accounted for a substantial share of variation in support; however, this should be interpreted with caution given that some covariates overlap with those used in the YCOM estimation.

4.4. Spatial patterns and model residuals

To further examine the geographic structure of support for the clean energy transition, we assessed both the spatial distribution of the raw outcome and the residual variation from the multilevel model. We first examined global spatial autocorrelation computing Global Moran's I using queen contiguity weights and 999 permutations, which indicated a strong positive spatial autocorrelation (Moran's $I = 0.63$, $z = 59.54$, $p < .001$): counties with similar levels of support were highly clustered in space. A LISA analysis (Fig. E) localized this clustering. High-high clusters, where counties with strong support were surrounded by similarly supportive counties, concentrated along the West Coast and in the Northeast. Low-low clusters, where counties with weaker support were surrounded by similarly low-support counties, were primarily located

across the Great Plains.

Residual analysis from the multilevel model indicated substantially weaker spatial dependence (Moran's $I = 0.12$, $z = 11.67$, $p = .001$), with the LISA results (Fig. F), showing markedly reduced and more fragmented residual clustering. Taken together, these diagnostics suggest that a large share of the geographic structure was captured by the included predictors. This reduction should be interpreted with caution, as some predictors are related through the YCOM estimation model. Even with this caveat, the broader pattern is clear: support for the clean energy transition is strongly spatially clustered across U.S. counties, and much of this clustering aligns with regional differences in risk perception and political context.

4.5. Robustness checks

We assessed model validity using standard diagnostics applied to Model 4. Residual versus fitted value plots (Fig. G.1) revealed no evidence of nonlinearity or systematic patterning, confirming the linearity assumption. Residuals were distributed symmetrically around zero with no meaningful heteroscedasticity. Normal Q-Q plots of standardized residuals (Fig. G.2) indicated approximate normality; minor tail deviations were negligible in magnitude and consistent with expectations for a large sample. Detrended Q-Q plots corroborated that these departures were inconsequential. Cook's distance statistics identified no influential outliers.

4.6. Support for the clean energy transition across political orientations

Stratified results revealed heterogeneity across political contexts despite the marginal or non-significant interaction term in the previous models (Tables E1–E3). Perceived climate risk was the strongest predictor of support for clean energy transition across all groups (all $p < .001$), with the association strongest in liberal ($\beta = 0.858$) and moderate counties ($\beta = 0.811$) and somewhat lower in conservative counties ($\beta = 0.556$), suggesting that risk perception is associated with policy support in politically more liberal communities.

Climate risk was positively associated with support in moderate and liberal counties in baseline models but became non-significant once perceived risk was included, consistent with perceived risk accounting for the observed association. In conservative counties, climate risk was non-significant across all specifications, converging with RQ1 findings: the association between objective climate risk and policy support varies systematically with political context.

Social vulnerability was positively associated with support across all models in conservative counties ($\beta = 0.217$ to 0.122). In liberal and moderate counties, the association was consistently negative.

5. Discussion

Support for the clean energy transition varies significantly across U.S. regions, reflecting local political and economic contexts as well as geographic patterns of climate and energy attitudes [6]. Our study examines geographic co-variation in how objective climate risks align with perceived risks and how both relate to support for a clean-energy transition across U.S. counties. The findings reveal a clear disconnect: objective climate risk is not consistently associated with support, while perceived risk aligns much more closely with policy preferences, with variation across local contexts, consistent with prior work [58,69,70,72,75,80,81].

Objective climate risk shows a small but significant positive association with support for clean energy transition, challenging assumptions that hazards map directly onto policy preferences [14,15] and echoing evidence that direct weather exposure has limited effects on beliefs and policy preferences [34]. Consistent with risk governance scholarship, objective risks do not automatically translate into recognition or action but must be interpreted within broader social and political systems [33,64,82]. This is most clearly illustrated in conservative counties, where the relationship reverses: *higher* objective risk is associated with *lower* perceived risk, and objective risk is not significantly associated with support for clean energy policies. Rather than reflecting a simple absence of risk responsiveness, this pattern suggests that in contexts characterized by lower institutional trust in climate science, the association between objective hazard exposure and perceived risk reverses, consistent with descriptions in the literature of resistance to threat recognition in politically polarized settings [64]. While Republicans remain less likely to acknowledge climate change [58,80], recent work suggests that local extreme weather can sometimes disrupt partisan biases in risk perception [83], and that perceived extreme weather events may shift risk perceptions more effectively among conservatives than liberals [15].

Perceived climate risk shows the strongest geographic alignment with support for clean-energy transition across all political contexts, though its strength varies, underscoring the centrality of ideology in a polarized landscape [29,72]. The association is strongest in liberal counties ($\beta = 0.858$) and weakest in conservative counties ($\beta = 0.556$). This variation is consistent with research on political polarization, suggesting that political orientation does not merely suppress climate concern but also limits the extent to which risk perception translates into policy support [68]. The observed association between perceived risk and support is therefore not uniformly available as a communication lever but is more constrained in politically resistant communities, a finding with direct implications for risk governance strategies that assume perceived risk reliably motivates policy support [65].

The social vulnerability findings reveal another interesting pattern. In the pooled models, social vulnerability shows a small negative association with support, masking opposing dynamic seen in stratified analyses. In liberal and moderate counties, greater social vulnerability is negatively associated with policy support, potentially reflecting that existing mitigation infrastructure may reduce the urgency of further climate action [14,15,50], while economic constraints may orient the vulnerable communities toward immediate material needs over longer-horizon climate policy priorities. In conservative counties by contrast, the association is positive. These county-level patterns may reflect differences in local mitigation infrastructure, institutional safety nets, clean energy transition as more urgent and personally relevant. These patterns might suggest that where vulnerability is experienced as unmitigated exposure, it may motivate support; where it is buffered by existing protective systems, it may not [14,15,50], though we cannot directly test the mechanisms underlying these patterns. It is also possible that this pattern partly reflects how YCOM-derived estimates perform differently across political subgroups, and this should be considered in interpretation.

Together, these divergent patterns suggest that social vulnerability

does not operate uniformly as either a barrier or a motivator for clean energy support but varies in direction and magnitude across political contexts in ways that challenge aggregate assumptions about how vulnerable communities relate to climate policy. What does remain consistent across all political contexts, however, is the central role of perceived risk. Taken together, these findings indicate that although ideology structures climate attitudes, perceived risk shows the strongest geographic alignment with policy support across all political contexts. Policy and communication strategies that make local climate risks concrete and salient, while addressing equity and access, are well-positioned to strengthen county-level support for the clean energy transition, particularly amid shifting national policy salience.

5.1. Policy recommendations

Building durable support for the clean energy transition in a polarized context requires aligning communication, equity, and coalition strategies with how people perceive risk and identify politically. The recommendations below translate our findings into actionable guidance.

1. Differentiate strategies by political context, not risk level.

The observed paradox, higher objective risk but lower perceived risk in conservative counties, signals a communication failure. In these contexts, conventional risk framing is unlikely to mobilize support. Messaging should instead emphasize locally salient priorities: infrastructure reliability, disaster preparedness, economic stability, and public health, grounded in lived experience rather than abstract data [54]. Highlighting firsthand accounts of extreme weather may be particularly effective, given evidence that experiential processing can shift attitudes even among politically resistant groups [83]. Trusted local messengers, farmers, small business owners, emergency managers, and faith leaders, are likely to outperform institutional voices where distrust of climate science runs high [1]. Across all audiences, efforts should connect local vulnerabilities explicitly to climate change, with targeted outreach to younger cohorts [15].

2. Prioritize perceived risk as a leverage point.

Perceived climate risk shows the strongest geographic alignment with support for clean-energy transition across all contexts, making it a primary target for engagement. However, gains may be constrained in politically resistant counties. Strategies should therefore center on place-based narratives linking recent local extreme weather to broader climate trends, participatory risk assessment, and framing that emphasizes personally meaningful consequences, threats to health, household finances, and local economies [84,85]. Heightening perceived risk salience is especially important where objective climate data have failed to shift attitudes [15].

3. Design vulnerability-sensitive policies tailored to political context

The divergent relationships between social vulnerability and policy support indicate that uniform approaches will be ineffective. In conservative counties, where vulnerability is positively associated with support, policies foregrounding concrete co-benefits, disaster resilience, energy cost stability, and job creation may resonate even in resistant environments. In moderate and liberal counties, where vulnerability is negatively associated with support, policy design should prioritize affordability and distributional equity, concerns that may otherwise suppress support among those most affected. Targeted instruments, on-bill financing, rate protections, community ownership models, and direct rebates for low-income households are particularly relevant. Ensuring equitable outcomes through targeted investment and inclusive planning strengthens both participation and long-term political durability [66].

4. Leverage state-level context in implementation

With over 40% of county-level variation in support attributable to state-level factors ($ICC = 0.416$), policy environments, regulatory structures, and political leadership at the state level play a decisive

role. Effective implementation strategies should therefore engage state institutions, utility commissions, legislative bodies, and regulatory agencies directly. Taken together, these recommendations leverage perceived risk, align with community priorities, and broaden coalitions, offering practical pathways toward an equitable and widely supported clean energy transition.

5.2. Limitations and future research

Several limitations should be considered when interpreting these findings. First, the estimation process in the YCOM data introduces potential dependencies among key variables. Both the dependent variable (support for clean energy transition) and a key independent variable (perceived climate risk) are county-level estimates derived from a multilevel regression and poststratification model that uses overlapping socioeconomic and demographic inputs, including Democrat vote share, same-sex households, and drive-alone commuting rates. Political orientation, the dominant predictor in our models, shares direct overlap with YCOM estimation. Consequently, the strong associations observed among political orientation, perceived risk, and policy support cannot be interpreted as fully independent empirical relationships as it partially reflects shared model inputs. Although sensitivity analyses indicate that the main patterns are not solely artifacts of this overlap, the extent to which results capture independent associations versus correlated estimation remains uncertain. Accordingly, these findings are best understood as descriptive of geographic co-variation among county-level objective risk, estimated attitudes, and political context, rather than as evidence of causal effects. Future research would benefit from incorporating independently measured attitudinal data, experimental designs, or longitudinal approaches to better isolate causal mechanisms and validate these relationships.

Second, the cross-sectional design identifies associations rather than causal relationships. Longitudinal designs are needed to establish temporal ordering and track how risk perceptions and policy attitudes evolve in response to extreme events, political discourse, or policy change. Third, aggregation to the county level masks within-county heterogeneity, and extending findings to individuals risks ecological fallacy. Fourth, the FEMA National Risk Index captures expected annual economic losses from natural hazards with composite indicators built from historical data and may underrepresent emerging risks. Finally, findings are specific to the United States and reflect a distinctive context of high political polarization around climate issues; the relationships observed here may not transfer to other contexts.

Future research should prioritize designs that address the measurement limitations identified here. Most directly, linking individual-level survey data to county- and state-level contextual indicators within a multilevel framework would allow independent measurement of perceived risk and policy support, thereby resolving the construct overlap that limits causal interpretation in the present study. Longitudinal panel designs paired with quasi-experimental approaches, including difference-in-differences, event studies, and randomized risk-communication trials, would help establish temporal ordering and identify mechanisms linking risk exposure, perception, and policy support across partisan contexts. Risk measurement should be broadened to include timely, multidimensional indicators beyond expected annual loss, such as PM_{2.5} and NO₂ exposure, heat stress, grid reliability, and service disruptions, ideally at higher temporal and spatial resolution. Real-time or high-frequency attitude tracking could capture shifts tied to extreme weather events, policy announcements, or media cycles that cross-sectional designs cannot detect. Community-engaged and equity-centered qualitative approaches would complement these quantitative designs by surfacing affordability, displacement, and access concerns among marginalized communities, improving both the relevance and justice dimensions of clean energy research. Cross-national comparisons can further test how institutional capacity, political alignment, media systems, and welfare regimes condition the relationship between risk

perception and policy support, sharpening strategies to build durable and equitable backing for the clean energy transition.

6. Conclusions

The findings of this study do not align with the assumption that mounting physical evidence will gradually bring public attitudes into line with scientific consensus, or that exposure, made vivid by intensifying hazards, will close the gap between what is known and what is supported. The communities most exposed to climate hazards are not consistently those where residents are most likely to perceive risk or to support a clean energy transition. The distance between exposure and support varies systematically with political identity and is paired with social vulnerability in different directions across contexts. Exposure, in other words, co-varies with public support only through the local conditions under which climate risk becomes visible and politically meaningful.

This points toward a different research and policy posture than the assumption that vivid exposure will close the gap on its own. If support varies with the local political and social conditions surrounding hazard, the central questions become contextual. Which political, institutional, and social arrangements coincide with closer alignment between perceived risk and exposure? Where does shared vulnerability co-occur with collective support rather than with division? And how do these conditions themselves shift as climate impacts intensify and political coalitions realign? These are empirical questions, answerable through the longitudinal, comparative, and community-engaged work that the cross-sectional design used here can motivate but not deliver.

Whether the patterns documented here persist as climate impacts intensify, and whether they hold in other countries where polarization around climate is shaped by different institutional and political conditions, are questions this study can raise but not settle. The relationship between climate risk and public support is unlikely to remain fixed. It will shift with the politics of the moment, the salience of recent events, and the institutions that stand between the two. Following how this relationship changes over time and across contexts will be essential for understanding when, where, and for whom durable public support for the clean energy transition takes shape.

CRedit authorship contribution statement

Lyudmyla Tsykalova: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Praise Ayotola Ayinla:** Writing – review & editing, Writing – original draft, Data curation. **Chien-Fei Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly and ChatGPT-4o to assist with language refinement and clarity in the preparation of this manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. These tools were not used for data analysis, interpretation, or original content generation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.erss.2026.104827>.

Data availability

Data will be made available on request.

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